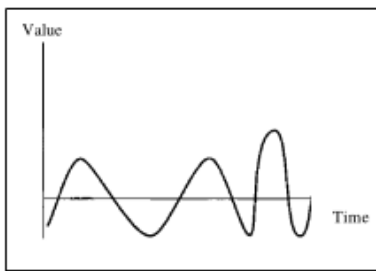


↓ ↓ What we have Studied in 3rd Chapter ↓ ↓

↓ ↓ Analog Signals (According to Book) ↓ ↓

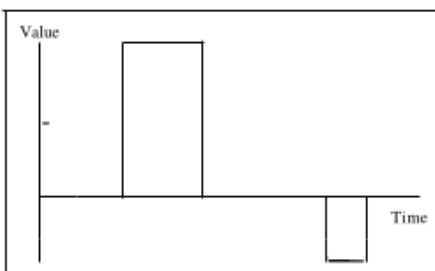
Analog signal has infinitely many levels of intensity over a period of time.



a. Analog signal

↓ ↓ Digital Signals (According to Book) ↓ ↓

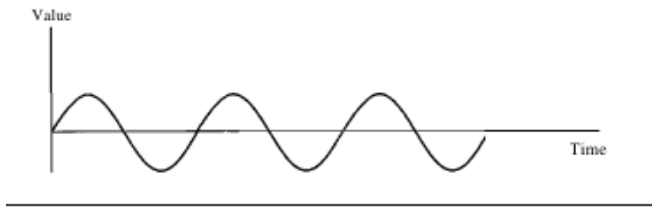
A digital signal, on the other hand, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.



b. Digital signal

↓ ↓ Periodic Signals ↓ ↓

A periodic signal completes a pattern within a measurable time frame, called a period, and repeats that pattern over subsequent identical periods.

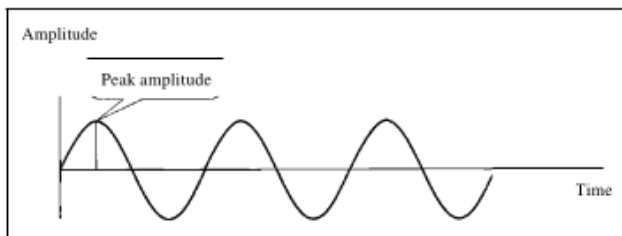


⇓ Non-Periodic Signals ⇓

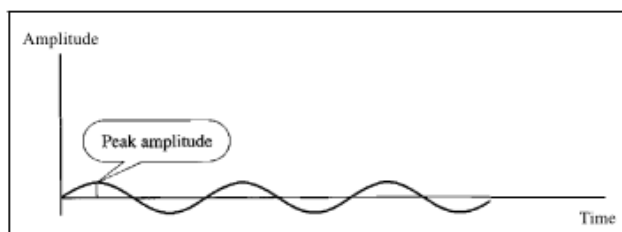
A nonperiodic signal changes without exhibiting a pattern or cycle that repeats over time.

⇓ Peak Amplitude ⇓

The peak amplitude of a signal is the absolute value of its highest intensity, proportional to the energy it carries. For electric signals, peak amplitude is normally measured in volts.



a. A signal with high peak amplitude

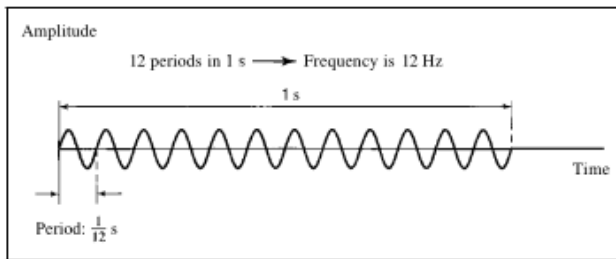


b. A signal with low peak amplitude

⇓ Period ⇓

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

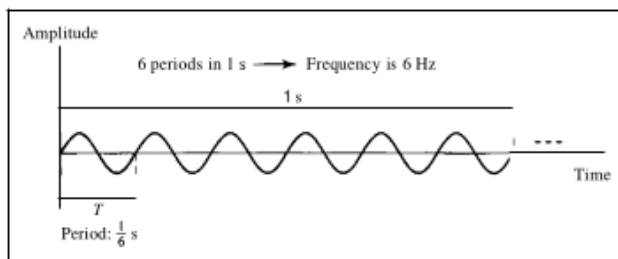
$$T = \frac{1}{f}$$



Frequency

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

$$f = \frac{1}{T}$$

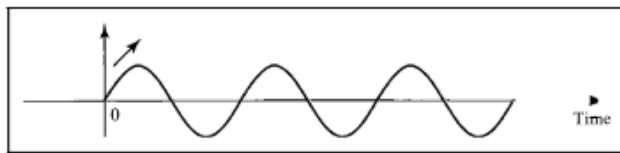


Units of Period and Frequency

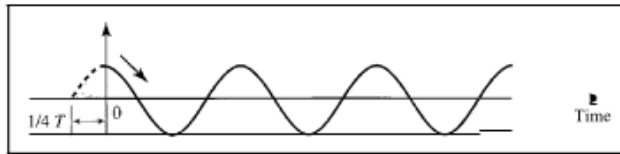
Unit	Equivalent	Unit	Equivalent
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^3 Hz
Microseconds (μ s)	10^{-6} s	Megahertz (MHz)	10^6 Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^9 Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Phase

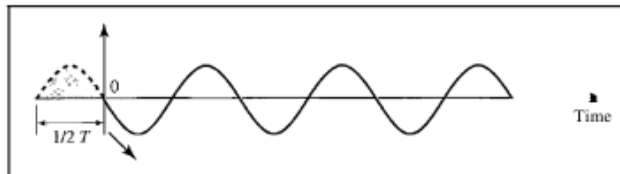
The term phase describes the position of the waveform relative to time 0. Phase is measured in degrees or radians [360° is 2π rad; 1° is $2\pi/360$ rad, and 1 rad is $360/(2\pi)$].



a. 0 degrees



b. 90 degrees



c. 180 degrees

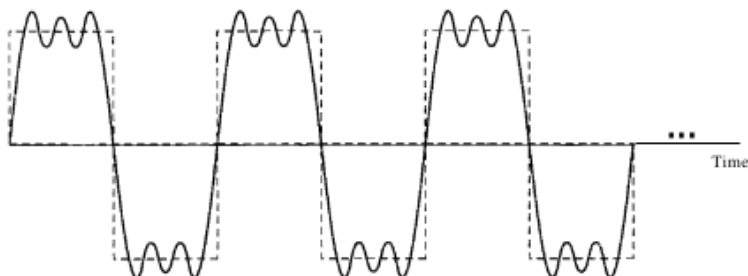
⇓ Wavelength ⇓

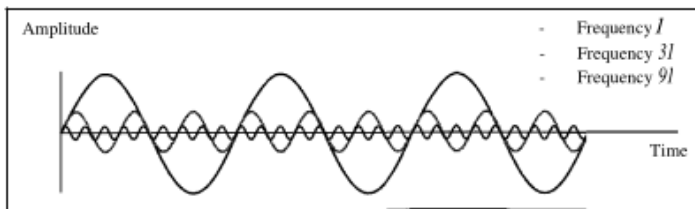
The distance one cycle occupies on the transmission medium.

$$Wavelength = \text{propagation speed} \times \text{period} = \frac{\text{propagation speed}}{\text{frequency}}$$

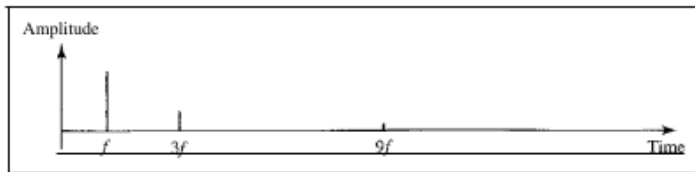
⇓ Composite Signals ⇓

Composite signals are combination of simple sine waves with different frequencies, amplitudes, and phases.





a. Time-domain decomposition of a composite signal

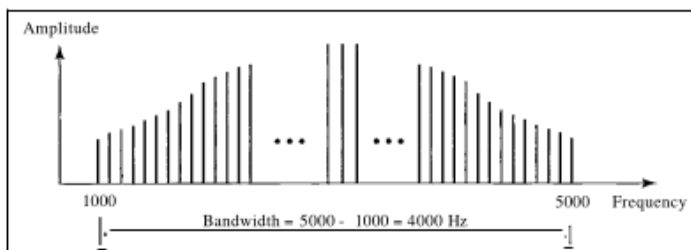


b. Frequency-domain decomposition of the composite signal

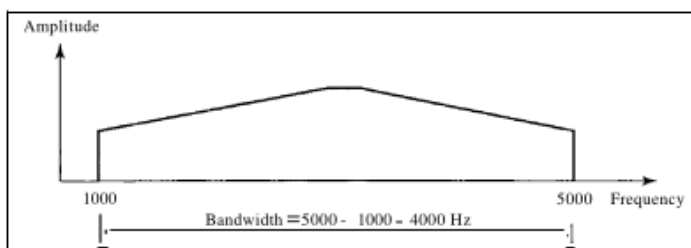
⇓ Bandwidth ⇓

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

$$B = f_h - f_l$$



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

⇓ Bit Rate ⇓

The bit rate is the number of bits sent in 1s, expressed in bits per second (bps).

$$\text{Bitrate} = \text{Pages per Minute} \times \text{Lines per Page} \times \text{Characters per Line} \times \text{Bits per Character}$$

⇓ Bit Length ⇓

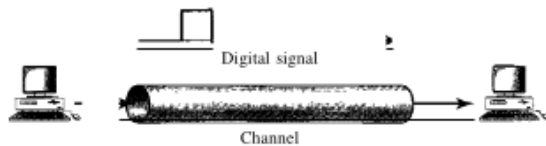
The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bitlength} = \text{propagation speed} \times \text{bit duration}$$

⇓ Baseband Transmission ⇓

Baseband transmission means sending a digital signal over a channel without changing the digital signal to an analog signal.

Figure 3.18 Baseband transmission



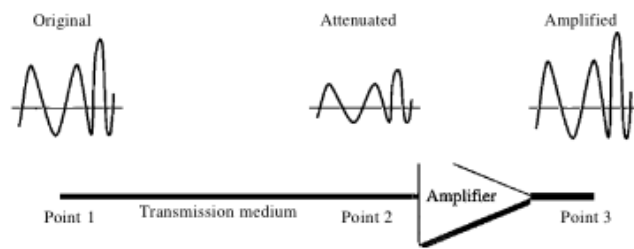
⇓ Broadband Transmission (Using Modulation) ⇓

Broadband transmission or modulation means changing the digital signal to an analog signal for transmission.

⇓ Attenuation ⇓

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.

Figure 3.26 Attenuation



⇓ Decibel ⇓

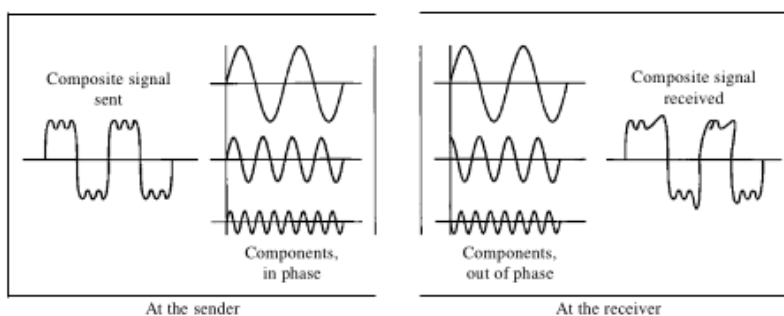
To show that a signal has lost or gained strength, engineers use the unit of the decibel. The decibel (dB) measures the relative strengths of two signals or one signal at two different points. Note that the decibel is negative if a signal is attenuated and positive if a signal is amplified.

$$dB = 10 \log_{10} \frac{P_2}{P_1}$$

⇓ Distortion ⇓

Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies. Each signal component has its own propagation speed through a medium and, therefore, its own delay in arriving at the final destination.

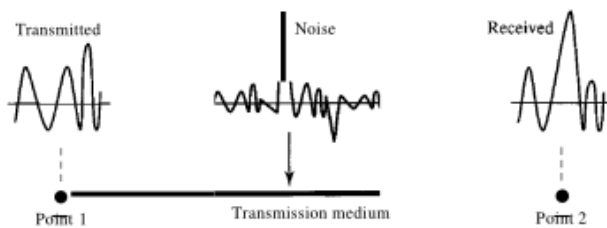
Figure 3.28 Distortion



↓ Noise ↓

Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal. **Thermal noise** is the random motion of electrons in a wire which creates an extra signal not originally sent by the transmitter. **Induced noise** comes from sources such as motors and appliances. **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna. Impulse noise is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

Figure 3.29 Noise



↓ Signal-to-Noise Ratio (SNR) ↓

As we know that, to find the theoretical bit rate limit, we need to know the ratio of the signal power to the noise power. The signal-to-noise ratio is defined as

$$SNR = \frac{\text{average signal power}}{\text{average noise power}}$$

Because **SNR** is the ratio of two powers, it is often described in decibel units,

$$SNR_{dB}$$

defined as

$$SNR_{dB} = 10 \log_{10} SNR$$

↓ Noiseless Channel: Nyquist Bit Rate ↓

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

$$BitRate = 2 \times bandwidth \times \log_2 L$$

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

⇓ Noisy Channel: Shannon Capacity ⇓

In reality, we cannot have a noiseless channel; the channel is always noisy. In 1944, Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

$$Capacity = bandwidth \times \log_2(1 + SNR)$$

⇓ Throughput ⇓

The throughput is a measure of how fast we can actually send data through a network.

⇓ Propagation Time ⇓

Propagation time measures the time required for a bit to travel from the source to the destination.

$$Propagationtime = \frac{Distance}{PropagationSpeed}$$

⇓ Latency (Delay) ⇓

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

$$Latency = \text{propagation time} + \text{transmission time} + \text{queuing time} + \text{processing delay}$$

↓ Transmission Time ↓

The time required for transmission of a message.

$$\text{Transmission time} = \frac{\text{MessageSize}}{\text{Bandwidth}}$$

↓ Jitter ↓

Jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example).

If the delay for the first packet is 20 ms, for the second is 45 ms, and for the third is 40 ms, then the real-time application that uses the packets endures jitter.

